



# Energy Skate Park

Location: <https://phet.colorado.edu/en/simulation/energy-skate-park>

## Part A: Intro

1. Go to the simulation at the URL above. It will give you a choice of 4 sections to click on (Intro, Measure, Graphs, and Playground). Start by clicking on Intro.
2. Click and drag on the skateboarder to move her to the top of the track. Let go, and observe her go back and forth.
3. Try out the slow motion feature. Click on *Slow* next to the play/pause button at the bottom. You can answer any of the following questions at full or slow speed.
4. Try out the *step* feature. Pause the simulation and hit the *step* button  to the right of the *pause/play* button to step forward one frame of animation at a time.
5. On the right side, you will see that Friction is set to “None.” Keep it that way for now...
6. What do you notice about the maximum height that the skater reaches on the left side compared to the maximum height on the right side? (Note: if you look in the bottom left of the simulation, you can turn on a *Grid* that will help to answer this, or you can use the tape measure  found in the bottom right corner.)

7. In the top right, turn on *Pie Chart*. A Pie Chart displaying a visual of the different form of energy in the skater-track-earth system will now follow your skater around.
  - a) At what point or points in the skater’s motion is kinetic energy at a maximum?

- b) At what point or points in the skater’s motion is potential energy at a maximum?

8. Turn on *Speed*. A Speedometer will now appear.

a) When the skater reaches the uppermost position(s), what is the speed value?

b) Multiple Choice: At what point is the speed at a maximum? .

A. Just before the skater reaches the very bottom

B. At the exact bottom

C. A little bit after reaching the bottom

9. Based on your previous observations, in what way is Speed related to Kinetic Energy?

10. Turn on *Path*. Dots will now trail your skater in the style of a motion diagram. Equal time intervals exist between each new dot's creation.

a) When are the dots furthest apart along the path? What does this indicate?

b) Explain how this makes sense in consideration of the equation for speed (speed = distance / time)

11. Experiment with the other three tracks . Try turning off the *Stick to Track* feature and see what happens on each.

12. On the left side of the simulation, click on  to turn on an animated Energy Bar Chart. Pause the simulation, and drag your skater up and down. What do you notice is happening to the amount of potential energy as you do this?

13. Generally speaking, in what way is Height related to Potential Energy?

14. With your skater paused in midair, adjust the Gravity slider back and forth between *Tiny* and *Lots*. Generally speaking, in what way is potential energy related to Gravity?

15. Turn on Reference Height (bottom left corner). A dotted line will appear with a pair of yellow arrows on it. You can raise and lower this dotted line. What can you do to make the amount of Potential Energy shown on the bar chart become *negative*?

16. Put your skater all the way to the top of the track and set the simulation into motion. Pause your simulation when the skater is about half-way up the curve so that both Kinetic Energy and Potential Energy exist on the Energy Bar Chart. Adjust the mass slider.

a) Generally speaking, in what way is Mass related to Potential Energy?

b) Generally speaking, in what way is Mass related to Kinetic Energy?

17. Set the skater into simple back and forth motion as before on the first track that is a simple U shape, and look at the Energy Bar Chart. Can you see how the rightmost bar labeled Total is remaining constant? The fact that it stays constant is a demonstration of the Law of...

18. Now look at the Kinetic and Potential bars. As Kinetic Energy increases, Potential Energy....

19. Try turning on friction and letting your skater start from the top of the track. Notice how the available energy is transforming into Thermal Energy.

- a) If you wait long enough, will the Thermal Energy transform back into Kinetic and Potential Energy?

- b) Will the skater eventually stop moving with the friction turned on? Describe what you observe about the amounts of the different forms of energy as time goes on.

- c) Try hitting the *trash*  icon below the Thermal bar. This will “dump out” the Thermal Energy. In reality this would represent the Thermal Energy dissipating into the environment surrounding the system.

20. There is one more way to increase Thermal Energy aside from using friction. Figure out what it is and describe it below.

## Part B: Measure

1. At the bottom of the screen, switch to the Measure section.
2. On either the 3rd or the 4th track  , bring your skater to the top left, and let your skater go. Keep *Stick to Track* turned on so that your skater cannot fall off the track or jump off the surface. Set friction to *none*.
3. Release your skater from the top left side of the track, and then pause when your skater reaches the top right side of the track. There will be a trail of dots visible on the track. You can click and drag the target of the data measuring tool onto different dots. It will then display the values of the different forms of energy as well as the speed and height of the skater when they were at the position of the dot your target is over.
4. Choose 10 data points that are distributed around the track. Try to pick data points on the left, bottom, middle, right, top. In other words, a “random” sampling. Record the statistics of these data points in the chart below. Click on the cells to enter values.

| Data Point | Height (m) | Speed (m/s) | Potential Energy (J) | Kinetic Energy (J) |
|------------|------------|-------------|----------------------|--------------------|
| 1          |            |             |                      |                    |
| 2          |            |             |                      |                    |
| 3          |            |             |                      |                    |
| 4          |            |             |                      |                    |
| 5          |            |             |                      |                    |
| 6          |            |             |                      |                    |
| 7          |            |             |                      |                    |
| 8          |            |             |                      |                    |
| 9          |            |             |                      |                    |
| 10         |            |             |                      |                    |

5. Using <http://thephysicsaviary.com/Physics/Programs/Tools/Graphing/> , graph Potential Energy ( $y$ -axis) vs. Height ( $x$ -axis). Screenshot and paste your results below, or simply sketch the graph.



6. Describe the exact mathematical relationship between Height and Potential Energy in words.



7. Now graph Kinetic Energy ( $y$ -axis) vs. Speed ( $x$ -axis). Screenshot and paste your results below, or simply sketch the graph.

8. Describe the exact mathematical relationship between Speed and Kinetic Energy in words.

9. Switch to track 2 . Set skater mass to 5 kg. With friction off, drop your skater from the uppermost track position, and measure the speed just after the end of the track.

10. Repeat this procedure using a skater mass of about 50 kg, and measure the speed just after the end of the track again. Are the results significantly different, or basically the same?

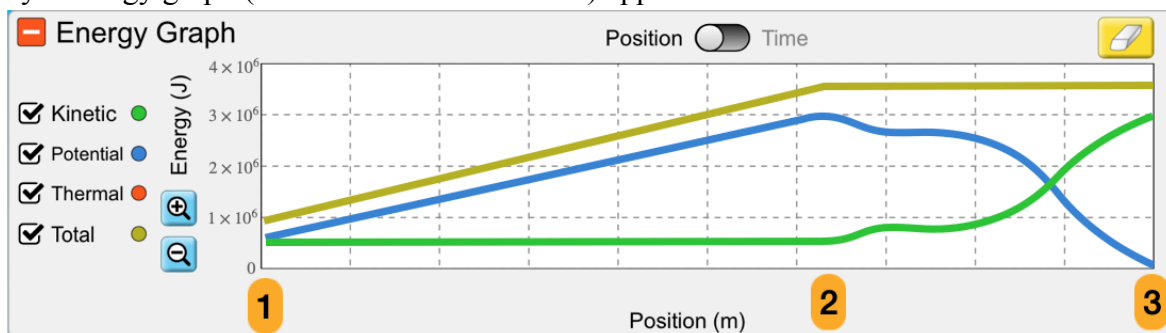
11. Just to make sure you aren't imagining things, try the experiment one more time with skater mass set to 100 kg. What do you notice about final speed relative to the previous two answers?

## Part C: Graphs

1. At the bottom of the screen, switch to the Graphs section. Take a few minutes to fully explore this section of the simulation. Click or tap on anything you can, slide all the sliders, and *Carefully* study the graphs that are produced as your skater moves around on either of the two tracks.



2. This is a picture of the Raging Bull ride at Six Flags Great America in Gurnee, IL. A section of the track is highlighted in yellow for our analysis. From point 1 to point 2, the cars are being pulled upward at a constant low speed. From point 2 to point 3 the carts are rolling down the track under the influence of gravity. Assume friction is negligible. A PhET style energy graph (with 1 intentional mistake) appears below:



- a) How can the total energy be increasing from point 1 to point 2? Where is that energy coming from?

- b) How come the Total Energy is greater than the potential energy from point 1 to point 2?



c) How come the Kinetic Energy is constant between points 1 and 2?



d) At what point in the graph do you notice there is a mistake? Look for something that is physically impossible, describe what you see and explain how you know that it is wrong.



## Part D: Playground

1. At the bottom of the screen, switch to the Playground section. Create your own course and either draw it below, or screenshot/paste it!